

Esercitazione RTL 2

Traccia del 12/04/2016

```
IRx → MAR, 0 → A;    (μ1)
M[MAR] → MBR, INCR(MAR) → MAR;    (μ2)
MBR → T1;    (μ3)
SHR(T1) → T1, T1 → T2;    (μ4)
c1: if OR(T1) == 1 then
|   M[MAR] → MBR, INCR(MAR) → MAR;    (μ2)
|   if MBR0 == 0 then
|   |   MBR → B;    (μ5)
|   |   A + B → A, DECR(T1) → T1, DECR(T2) →
|   |   |   T2, goto c1;    (μ6)
|   |   else
|   |   |   DECR(T1) → T1, DECR(T2) →
|   |   |   |   T2, goto c1;    (μ7)
|   |   fi
|   fi
|   else
c2: if OR(T2) == 1 then
|   M[MAR] → MBR, INCR(MAR) →
|   |   MAR;    (μ2)
|   |   if MBR0 == 1 then
|   |   |   MBR → B;    (μ5)
|   |   |   A - B → A, DECR(T2) → T2, goto c2;    (μ8)
|   |   |   else
|   |   |   |   DECR(T2) → T2, goto c2;    (μ9)
|   |   |   fi
|   |   fi
|   |   else
|   |   |   A → AC;    (μ10)
|   |   |   fi
|   fi
fi
```

```
IRx → MAR, 0 → A;    (μ1)
M[MAR] → MBR, INCR(MAR) → MAR;    (μ2)
MBR → T1;    (μ3)
SHR(T1) → T1;    (μ4)
T1 → T2;    (μ5)
c1: if OR(T1) == 1 then
|   M[MAR] → MBR, INCR(MAR) → MAR;    (μ2)
|   if MBR0 == 0 then
|   |   MBR → B;    (μ6)
|   |   A + B → A, DECR(T1) → T1, DECR(T2) →
|   |   |   T2, goto c1;    (μ7)
|   |   else
|   |   |   DECR(T1) → T1, DECR(T2) →
|   |   |   |   T2, goto c1;    (μ8)
|   |   |   fi
|   |   fi
|   else
c2: if OR(T2) == 1 then
|   M[MAR] → MBR, INCR(MAR) →
|   |   MAR;    (μ2)
|   |   if MBR0 == 1 then
|   |   |   MBR → B;    (μ5)
|   |   |   A - B → A, DECR(T2) → T2, goto c2;    (μ9)
|   |   |   else
|   |   |   |   DECR(T2) → T2, goto c2;    (μ10)
|   |   |   |   fi
|   |   |   fi
|   |   else
|   |   |   A → AC;    (μ11)
|   |   |   fi
|   fi
fi
```

```
IRx → MAR, 0 → A;    (μ1)
M[MAR] → MBR, INCR(MAR) → MAR;    (μ2)
MBR → T1;    (μ3)
SHR(T1) → T1, T1 → T2;    (μ4)
c1: if OR(T1) == 1 then
|   M[MAR] → MBR, INCR(MAR) → MAR;    (μ2)
|   if MBR0 == 0 then
|   |   MBR → B;    (μ5)
|   |   A + B → A, DECR(T1) → T1, DECR(T2) →
|   |   |   T2, goto c1;    (μ6)
|   |   else
|   |   |   DECR(T1) → T1, DECR(T2) →
|   |   |   |   T2, goto c1;    (μ7)
|   |   |   fi
|   |   fi
|   else
c2: if OR(T2) == 1 then
|   M[MAR] → MBR, INCR(MAR) →
|   |   MAR;    (μ2)
|   |   if MBR0 == 1 then
|   |   |   DECR(T2) → T2;    (μ8)
|   |   |   MBR → B, A - B → A, goto c2;    (μ9)
|   |   |   else
|   |   |   |   DECR(T2) → T2, goto c2;    (μ10)
|   |   |   |   fi
|   |   |   fi
|   |   else
|   |   |   A → AC;    (μ11)
|   |   |   fi
|   fi
fi
```

Traccia del 07/07/11

```

       $IR_x \rightarrow MAR, 32 \rightarrow T_1, 0 \rightarrow B; \quad (\mu_1)$ 
c: if  $OR(T_1) == 1$  then
  |
  | if  $AC_0 == 1$  then
  | |  $M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow$ 
  | |  $MAR; \quad (\mu_2)$ 
  | |  $MBR \rightarrow A; \quad (\mu_3)$ 
  | |  $A + B \rightarrow B, DECR(T_1) \rightarrow T_1, SHR(AC) \rightarrow$ 
  | |  $AC, \textbf{goto } c; \quad (\mu_4)$ 
  | else
  | |  $INCR(MAR) \rightarrow MAR, DECR(T_1) \rightarrow$ 
  | |  $T_1, SHR(AC) \rightarrow AC, \textbf{goto } c; \quad (\mu_5)$ 
  | fi
else
  |  $B \rightarrow AC; \quad (\mu_6)$ 
fi
```

```

       $IR_x \rightarrow MAR, 32 \rightarrow T_1, 0 \rightarrow B; \quad (\mu_1)$ 
c: if  $OR(T_1) == 1$  then
  |
  | if  $AC_0 == 1$  then
  | |  $M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow$ 
  | |  $MAR; \quad (\mu_2)$ 
  | |  $MBR \rightarrow A, A + B \rightarrow B; \quad (\mu_3)$ 
  | |  $DECR(T_1) \rightarrow T_1, SHR(AC) \rightarrow$ 
  | |  $AC, \textbf{goto } c; \quad (\mu_4)$ 
  | else
  | |  $INCR(MAR) \rightarrow MAR, DECR(T_1) \rightarrow$ 
  | |  $T_1, SHR(AC) \rightarrow AC, \textbf{goto } c; \quad (\mu_5)$ 
  | fi
else
  |  $B \rightarrow AC; \quad (\mu_6)$ 
fi
```

```

       $IR_x \rightarrow MAR, 32 \rightarrow T_1, 0 \rightarrow B; \quad (\mu_1)$ 
c: if  $OR(T_1) == 1$  then
  |
  | if  $AC_0 == 1$  then
  | |  $M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow$ 
  | |  $MAR; \quad (\mu_2)$ 
  | |  $MBR \rightarrow A; \quad (\mu_3)$ 
  | |  $A + B \rightarrow B, DECR(T_1) \rightarrow T_1, SHR(AC) \rightarrow$ 
  | |  $AC, \textbf{goto } c; \quad (\mu_4)$ 
  | else
  | |  $SHR(AC) \rightarrow AC; \quad (\mu_5)$ 
  | |  $INCR(MAR) \rightarrow MAR, DECR(T_1) \rightarrow$ 
  | |  $T_1, \textbf{goto } c; \quad (\mu_6)$ 
  | fi
else
  |  $B \rightarrow AC; \quad (\mu_7)$ 
fi
```

Traccia del 16/09/2014

```
 $IR_x \rightarrow MAR;$      $(\mu_1)$   
 $INC(MAR) \rightarrow MAR;$      $(\mu_2)$   
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$      $(\mu_3)$   
 $MBR \rightarrow T1, M[MAR] \rightarrow MBR;$      $(\mu_4)$   
 $MBR \rightarrow MAR, 0 \rightarrow B;$      $(\mu_5)$   
c: if  $OR(T1) == 1$  then  
     $M[MAR] \rightarrow MBR, INC(MAR) \rightarrow MAR;$      $(\mu_6)$   
    if  $\beta_1 == 0$  then  
         $DEC(T1) \rightarrow T1, \text{goto } c;$      $(\mu_7)$   
    else  
         $MBR \rightarrow A;$      $(\mu_8)$   
         $A + B \rightarrow B, DEC(T1) \rightarrow T1, \text{goto } c;$      $(\mu_9)$   
    fi  
    else  
         $B \rightarrow ACC;$      $(\mu_{10})$   
    fi
```

```
 $IR_x \rightarrow MAR;$      $(\mu_1)$   
 $INC(MAR) \rightarrow MAR;$      $(\mu_2)$   
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$      $(\mu_3)$   
 $MBR \rightarrow T1, M[MAR] \rightarrow MBR;$      $(\mu_4)$   
 $MBR \rightarrow MAR, 0 \rightarrow B;$      $(\mu_5)$   
c: if  $OR(T1) == 1$  then  
     $M[MAR] \rightarrow MBR, INC(MAR) \rightarrow MAR;$      $(\mu_6)$   
    if  $\beta_1 == 0$  then  
         $DEC(T1) \rightarrow T1, \text{goto } c;$      $(\mu_7)$   
    else  
         $MBR \rightarrow A, DEC(T1) \rightarrow T1;$      $(\mu_8)$   
         $A + B \rightarrow B, \text{goto } c;$      $(\mu_9)$   
    fi  
    else  
         $B \rightarrow ACC;$      $(\mu_{10})$   
    fi
```

```
 $IR_x \rightarrow MAR;$      $(\mu_1)$   
 $INC(MAR) \rightarrow MAR;$      $(\mu_2)$   
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$      $(\mu_3)$   
 $MBR \rightarrow T1, M[MAR] \rightarrow MBR;$      $(\mu_4)$   
 $MBR \rightarrow MAR, 0 \rightarrow A;$      $(\mu_5)$   
c: if  $OR(T1) == 1$  then  
     $M[MAR] \rightarrow MBR, INC(MAR) \rightarrow MAR;$      $(\mu_6)$   
    if  $\beta_1 == 0$  then  
         $DEC(T1) \rightarrow T1, \text{goto } c;$      $(\mu_7)$   
    else  
         $MBR \rightarrow B;$      $(\mu_8)$   
         $A + B \rightarrow A, DEC(T1) \rightarrow T1, \text{goto } c;$      $(\mu_9)$   
    fi  
    else  
         $A \rightarrow ACC;$      $(\mu_{10})$   
    fi
```

Traccia del 30/04/2014

```
 $IR_x \rightarrow MAR, ACC \rightarrow T2; \quad (\mu_1)$   
 $M[MAR] \rightarrow MBR; \quad (\mu_2)$   
 $MBR \rightarrow T1; \quad (\mu_3)$   
c: if  $OR(T2) == 1$  then  
    if  $T1_0 == 1$  then  
         $0 \rightarrow ACC; \quad (\mu_4)$   
    else  
         $SHR(T1) \rightarrow T1, DEC(T2) \rightarrow T2, \textbf{goto c}; \quad (\mu_5)$   
    fi  
else  
     $1 \rightarrow ACC; \quad (\mu_6)$   
fi
```

```
 $IR_x \rightarrow MAR; \quad (\mu_1)$   
 $ACC \rightarrow T2; \quad (\mu_2)$   
 $M[MAR] \rightarrow MBR, MBR \rightarrow T1; \quad (\mu_3)$   
c: if  $OR(T2) == 1$  then  
    if  $T1_0 == 1$  then  
         $0 \rightarrow ACC; \quad (\mu_4)$   
    else  
         $SHR(T1) \rightarrow T1, DEC(T2) \rightarrow T2, \textbf{goto c}; \quad (\mu_5)$   
    fi  
else  
     $1 \rightarrow ACC; \quad (\mu_6)$   
fi
```

```
 $IR_x \rightarrow MAR; \quad (\mu_1)$   
 $M[MAR] \rightarrow MBR; \quad (\mu_2)$   
 $MBR \rightarrow T1, ACC \rightarrow T2; \quad (\mu_3)$   
c: if  $OR(T2) == 1$  then  
    if  $T1_0 == 1$  then  
         $0 \rightarrow ACC; \quad (\mu_4)$   
    else  
         $SHR(T1) \rightarrow T1, DEC(T2) \rightarrow T2, \textbf{goto c}; \quad (\mu_5)$   
    fi  
else  
     $1 \rightarrow ACC; \quad (\mu_6)$   
fi
```

Traccia del 19/03/2007

$IR_x \rightarrow MAR;$ (μ_1)
 $M[MAR] \rightarrow MBR, INC(MAR) \rightarrow MAR;$ (μ_2)
 $MBR \rightarrow B, M[MAR] \rightarrow MBR;$ (μ_3)
 $MBR \rightarrow A;$ (μ_4)
 $A - B \rightarrow T1;$ (μ_5)
if $T1_{31} == 0$ **then**
 if $OR(T1) == 0$ **then**
 $B \rightarrow MAR;$ (μ_6)
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$ (μ_7)
 else
 $B \rightarrow MAR;$ (μ_6)
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$ (μ_7)
 $SHR(MBR) \rightarrow MBR;$ (μ_8)
 $INC(MBR) \rightarrow MBR;$ (μ_9)
 fi
else
 $B \rightarrow MAR;$ (μ_6)
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$ (μ_7)
 $MBR \rightarrow B;$ (μ_{10})
 $A + B \rightarrow MBR;$ (μ_{11})
fi
 $MBR \rightarrow M[MAR];$ (μ_{12})

$IR_x \rightarrow MAR;$ (μ_1)
 $M[MAR] \rightarrow MBR, INC(MAR) \rightarrow MAR;$ (μ_2)
 $MBR \rightarrow A, M[MAR] \rightarrow MBR;$ (μ_3)
 $MBR \rightarrow B;$ (μ_4)
 $B - A \rightarrow T1;$ (μ_5)
if $T1_{31} == 0$ **then**
 if $OR(T1) == 0$ **then**
 $B \rightarrow MAR;$ (μ_6)
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$ (μ_7)
 else
 $B \rightarrow MAR;$ (μ_6)
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$ (μ_7)
 $SHR(MBR) \rightarrow MBR;$ (μ_8)
 $INC(MBR) \rightarrow MBR;$ (μ_9)
 fi
else
 $B \rightarrow MAR;$ (μ_6)
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$ (μ_7)
 $MBR \rightarrow B;$ (μ_{10})
 $A + B \rightarrow MBR;$ (μ_{11})
fi
 $MBR \rightarrow M[MAR];$ (μ_{12})

$IR_x \rightarrow MAR;$ (μ_1)
 $M[MAR] \rightarrow MBR, INC(MAR) \rightarrow MAR;$ (μ_2)
 $MBR \rightarrow B, M[MAR] \rightarrow MBR;$ (μ_3)
 $MBR \rightarrow A;$ (μ_4)
 $A - B \rightarrow T1;$ (μ_5)
if $T1_{31} == 0$ **then**
 if $OR(T1) == 0$ **then**
 $B \rightarrow MAR;$ (μ_6)
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$ (μ_7)
 else
 $B \rightarrow MAR;$ (μ_6)
 $M[MAR] \rightarrow MBR;$ (μ_8)
 $IR_x \rightarrow MAR;$ (μ_1)
 $SHR(MBR) \rightarrow MBR, INC(MBR) \rightarrow MBR;$ (μ_9)
 fi
else
 $B \rightarrow MAR;$ (μ_6)
 $M[MAR] \rightarrow MBR, IR_x \rightarrow MAR;$ (μ_7)
 $MBR \rightarrow B;$ (μ_{10})
 $A + B \rightarrow MBR;$ (μ_{11})
fi
 $MBR \rightarrow M[MAR];$ (μ_{12})

Traccia del 18/04/2007

```
 $\mu_1 : IR_x \rightarrow MAR$   
 $\mu_2 : M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR, 0 \rightarrow B$   
c: if  $OR(MBR) == 1$  then  
|  $\mu_3 : MBR \rightarrow A$   
| if  $MBR_0 == 0$  then  
| |  $\mu_4 : SHR(A) \rightarrow A, A + B \rightarrow B$   
| |  $\mu_5 : M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR, goto\ c$   
| else  
| |  $\mu_6 : SHL(A) \rightarrow A$   
| |  $\mu_5 : A + B \rightarrow B, M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR, goto\ c$   
| fi  
else  
|  $\mu_7 : B \rightarrow AC$   
fi
```

```
 $\mu_1 : IR_x \rightarrow MAR, 0 \rightarrow B$   
 $\mu_2 : M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR$   
c: if  $OR(MBR) == 1$  then  
|  $\mu_3 : MBR \rightarrow A$   
| if  $MBR_0 == 0$  then  
| |  $\mu_4 : SHR(A) \rightarrow A$   
| |  $\mu_5 : A + B \rightarrow B, M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR, goto\ c$   
| else  
| |  $\mu_6 : SHL(A) \rightarrow A$   
| |  $\mu_5 : A + B \rightarrow B, M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR, goto\ c$   
| fi  
else  
|  $\mu_7 : B \rightarrow AC$   
fi
```

```
 $\mu_1 : IR_x \rightarrow MAR, 0 \rightarrow B$   
 $\mu_2 : M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR$   
c: if  $OR(MBR) == 1$  then  
|  $\mu_3 : MBR \rightarrow A$   
| if  $MBR_0 == 0$  then  
| |  $\mu_4 : SHR(A) \rightarrow A$   
| |  $\mu_5 : A + B \rightarrow B$   
| |  $\mu_6 : M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR, goto\ c$   
| else  
| |  $\mu_7 : SHL(A) \rightarrow A, A + B \rightarrow B, M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR, goto\ c$   
| fi  
else  
|  $\mu_8 : B \rightarrow AC$   
fi
```

Traccia del 30/04/2019

```
 $\mu_1 : IR_x \rightarrow MAR, 0 \rightarrow AC$   
 $\mu_2 : M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR$   
 $\mu_3 : MBR \rightarrow T1, M[MAR] \rightarrow$   
     $MBR, INCR(MAR) \rightarrow MAR$   
 $\mu_4 : SHR(T1) \rightarrow T1$   
c: if  $OR(T1) == 0$  then  
     $\mu_4 : MBR \rightarrow A, M[MAR] \rightarrow$   
         $MBR, INCR(MAR) \rightarrow MAR$   
     $\mu_4 : MBR \rightarrow B, M[MAR] \rightarrow$   
         $MBR, INCR(MAR) \rightarrow MAR$   
     $\mu_5 : A + B \rightarrow A$   
     $\mu_6 : MBR \rightarrow B$   
     $\mu_7 : A - B \rightarrow B$   
    if  $B_{31} == 1$  then  
         $\mu_8 : INCR(AC) \rightarrow AC, DECR(T1) \rightarrow$   
             $T1, goto\ c$   
    else  
         $\mu_9 : DECR(T1) \rightarrow T1, goto\ c$   
    fi  
else  
     $\mu_0 : \phi$   
fi
```

```
 $\mu_1 : IR_x \rightarrow MAR$   
 $\mu_2 : M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR, 0 \rightarrow$   
     $AC$   
 $\mu_3 : MBR \rightarrow T1, M[MAR] \rightarrow$   
     $MBR, INCR(MAR) \rightarrow MAR$   
 $\mu_4 : SHR(T1) \rightarrow T1$   
c: if  $OR(T1) == 1$  then  
     $\mu_4 : MBR \rightarrow A, M[MAR] \rightarrow$   
         $MBR, INCR(MAR) \rightarrow MAR$   
     $\mu_4 : MBR \rightarrow B, M[MAR] \rightarrow$   
         $MBR, INCR(MAR) \rightarrow MAR$   
     $\mu_5 : A + B \rightarrow A$   
     $\mu_6 : MBR \rightarrow B$   
     $\mu_7 : A - B \rightarrow B$   
    if  $B_{31} == 1$  then  
         $\mu_8 : INCR(AC) \rightarrow AC, DECR(T1) \rightarrow$   
             $T1, goto\ c$   
    else  
         $\mu_9 : DECR(T1) \rightarrow T1, goto\ c$   
    fi  
else  
     $\mu_0 : \phi$   
fi
```

```
 $\mu_1 : IR_x \rightarrow MAR, 0 \rightarrow AC$   
 $\mu_2 : M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR$   
 $\mu_3 : MBR \rightarrow T1, M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR$   
 $\mu_4 : SHR(T1) \rightarrow T1$   
c: if  $OR(T1) == 0$  then  
     $\mu_4 : MBR \rightarrow A, M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow$   
         $MAR$   
     $\mu_4 : MBR \rightarrow B, M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow$   
         $MAR$   
     $\mu_5 : A + B \rightarrow A$   
     $\mu_6 : MBR \rightarrow B, A - B \rightarrow B$   
    if  $B_{31} == 1$  then  
         $\mu_7 : INCR(AC) \rightarrow AC, DECR(T1) \rightarrow T1, goto\ c$   
    else  
         $\mu_8 : DECR(T1) \rightarrow T1, goto\ c$   
    fi  
else  
     $\mu_0 : \phi$   
fi
```

Traccia del 31/03/2009

```
 $IR_x \rightarrow A, 16 \rightarrow T1; \quad (\mu_1)$   
 $IR_x \rightarrow IND1, 16 \rightarrow B; \quad (\mu_2)$   
 $A + B \rightarrow IND2; \quad (\mu_3)$   
c: if  $OR(T1) == 1$  then  
|  $IND1 \rightarrow MAR; \quad (\mu_4)$   
|  $M[MAR] \rightarrow MBR, IND2 \rightarrow MAR; \quad (\mu_5)$   
|  $MBR \rightarrow B, M[MAR] \rightarrow MBR; \quad (\mu_6)$   
|  $MBR \rightarrow A; \quad (\mu_7)$   
|  $A + B \rightarrow A; \quad (\mu_8)$   
| if  $OR(A) == 0$  then  
| |  $IND1 \rightarrow MAR; \quad (\mu_5)$   
| |  $MBR \rightarrow M[MAR], B \rightarrow MBR, IND2 \rightarrow MAR;$   
| |  $\quad (\mu_9)$   
| |  $MBR \rightarrow M[MAR], INC(IND1) \rightarrow$   
| |  $IND1, INC(IND2) \rightarrow IND2, DEC(T1) \rightarrow$   
| |  $T1, \mathbf{goto} \text{ c}; \quad (\mu_{10})$   
| else  
| |  $INC(IND1) \rightarrow IND1, INC(IND2) \rightarrow$   
| |  $IND2, DEC(T1) \rightarrow T1, \mathbf{goto} \text{ c}; \quad (\mu_{11})$   
| fi  
else  
|  $\phi; \quad (\mu_0)$   
fi
```

```
 $IR_x \rightarrow A, 16 \rightarrow B, 16 \rightarrow T1; \quad (\mu_1)$   
 $IR_x \rightarrow IND1; \quad (\mu_2)$   
 $A + B \rightarrow IND2; \quad (\mu_3)$   
c: if  $OR(T1) == 1$  then  
|  $IND1 \rightarrow MAR; \quad (\mu_4)$   
|  $M[MAR] \rightarrow MBR, IND2 \rightarrow MAR; \quad (\mu_5)$   
|  $MBR \rightarrow B, M[MAR] \rightarrow MBR; \quad (\mu_6)$   
|  $MBR \rightarrow A; \quad (\mu_7)$   
|  $A + B \rightarrow A; \quad (\mu_8)$   
| if  $OR(A) == 1$  then  
| |  $IND1 \rightarrow MAR; \quad (\mu_5)$   
| |  $MBR \rightarrow M[MAR], IND2 \rightarrow MAR; \quad (\mu_9)$   
| |  $B \rightarrow M[MAR], INC(IND1) \rightarrow$   
| |  $IND1, INC(IND2) \rightarrow IND2, DEC(T1) \rightarrow$   
| |  $T1, \mathbf{goto} \text{ c}; \quad (\mu_{10})$   
| else  
| |  $INC(IND1) \rightarrow IND1, INC(IND2) \rightarrow$   
| |  $IND2, DEC(T1) \rightarrow T1, \mathbf{goto} \text{ c}; \quad (\mu_{11})$   
| fi  
else  
|  $\phi; \quad (\mu_0)$   
fi
```

```
 $IR_x \rightarrow IND1, 16 \rightarrow B, 16 \rightarrow T1; \quad (\mu_1)$   
 $IR_x \rightarrow A; \quad (\mu_2)$   
 $A + B \rightarrow IND2; \quad (\mu_3)$   
c: if  $OR(T1) == 1$  then  
|  $IND1 \rightarrow MAR; \quad (\mu_4)$   
|  $M[MAR] \rightarrow MBR, IND2 \rightarrow MAR; \quad (\mu_5)$   
|  $MBR \rightarrow B, M[MAR] \rightarrow MBR; \quad (\mu_6)$   
|  $MBR \rightarrow A; \quad (\mu_7)$   
|  $A + B \rightarrow A; \quad (\mu_8)$   
| if  $OR(A) == 1$  then  
| |  $IND1 \rightarrow MAR; \quad (\mu_5)$   
| |  $MBR \rightarrow M[MAR], B \rightarrow MBR, IND2 \rightarrow MAR;$   
| |  $\quad (\mu_9)$   
| |  $MBR \rightarrow M[MAR], INC(IND1) \rightarrow$   
| |  $IND1, INC(IND2) \rightarrow IND2, DEC(T1) \rightarrow$   
| |  $T1, \mathbf{goto} \text{ c}; \quad (\mu_{10})$   
| else  
| |  $INC(IND1) \rightarrow IND1, INC(IND2) \rightarrow$   
| |  $IND2, DEC(T1) \rightarrow T1, \mathbf{goto} \text{ c}; \quad (\mu_{11})$   
| fi  
else  
|  $\phi; \quad (\mu_0)$   
fi
```